Bicol University

POLANGUI CAMPUS

Centro Occidental Polangui, Albay

Project Proposal in IT 112: Web Systems and Technologies
2nd Semester A.Y 2025 - 2026

NEXUSWrites
"Your Tech, Your Story."

Group Name:
Powerpuffgals

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Course & Block:
BSIT-2A

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Table of Contents

Cover Page…….…….…….……....1

Proponents……..……..…….…...2

Table of Contents……..…….…..……3

Project Proposal…….…….……4

Introduction……...4

System Purpose……..4

Target Users……...……4

Key Features…….……5

Use Case Diagram…….…..……...…...6

Entity Relationship Diagram (ERD)……..……..7

Data Flow Diagram ……..……..……..…..…...……..8

GitHub Repository URL & Screenshot of Repo Structure…….…….9

Signatories…….……..….……..…10

Final Project Proposal: NexusWrites - “Your Tech, Your Story.”

I. Introduction

NexusWrites is a specialized publishing platform designed for the IT community, where technical documentation meets professional networking. Unlike generic blogging sites, NexusWrites provides a structured environment for students and professionals to archive laboratory activities, document project workflows, and share verified technical expertise. By focusing on high-quality content—such as code-heavy tutorials and technical reviews—the platform filters out the "noise" of social media, creating a credible space for professional identity management and technical growth.

II. System Purpose

The primary goal of **NexusWrites** is to provide a high-performance environment for structured knowledge sharing. The system bridges the gap between academic tutorials and real-world development through:

- **Knowledge Exchange:** A specialized workspace supporting rich-text documentation, code snippets, and integrated technical imagery.
- **Professional Networking:** A structured interaction hub that facilitates collaborative problem-solving through a threaded discussion system.
- **Industry Integration:** A bridge to professional practice, allowing users to link their online content to their real-world code repositories and technical milestones.

III. Target Users

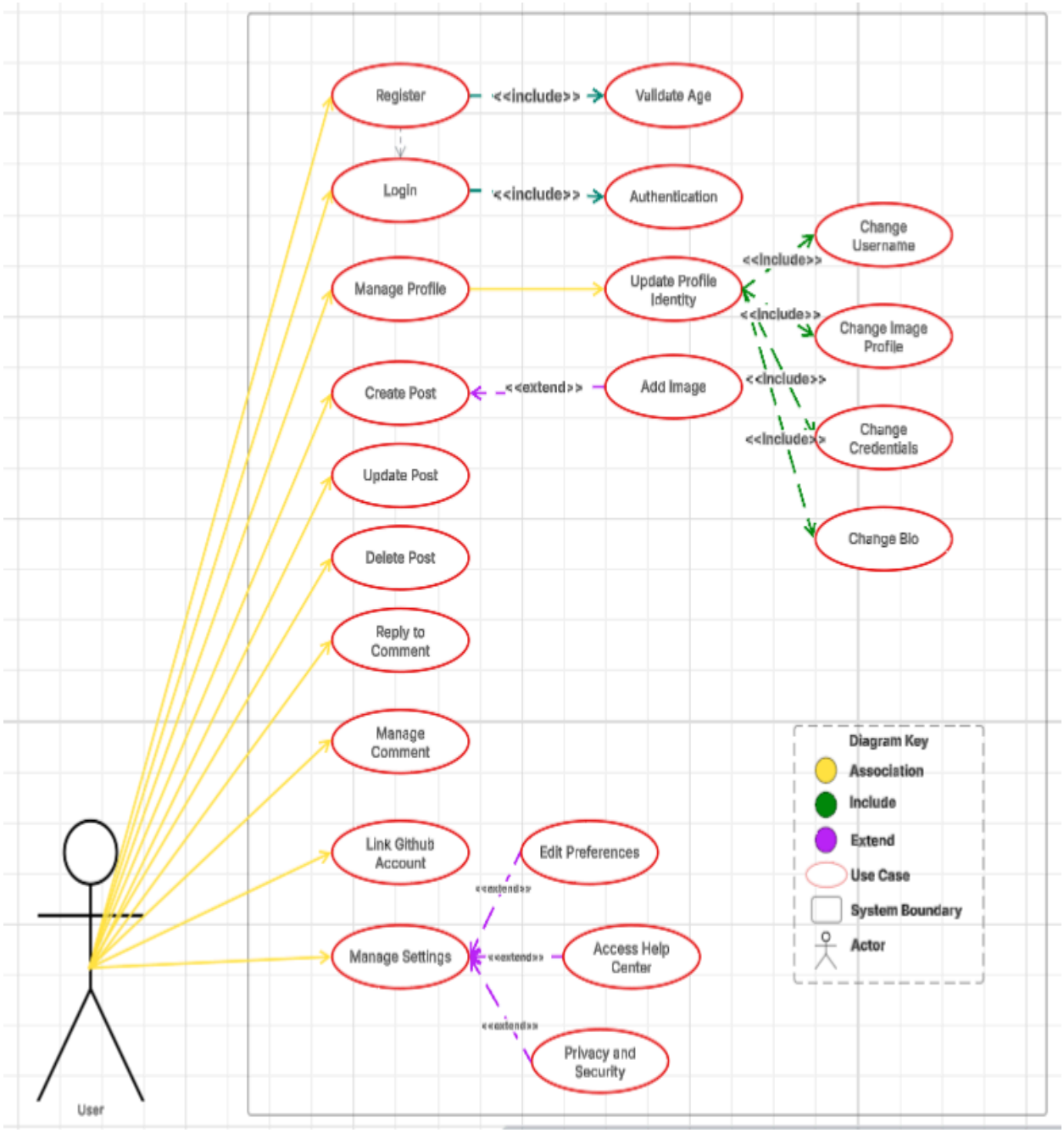
- **Technology Students:** Undergraduate students (BSIT, BSCS, etc.) seeking a professional space to document laboratory activities and projects.
- **Tech Professionals:** Developers and engineers sharing technical insights, mentoring peers, and archiving tutorials on modern tools.
- **General Tech Enthusiasts:** Individuals interested in learning about coding, hardware, or AI trends through high-level technical discussions.

IV. Key Features

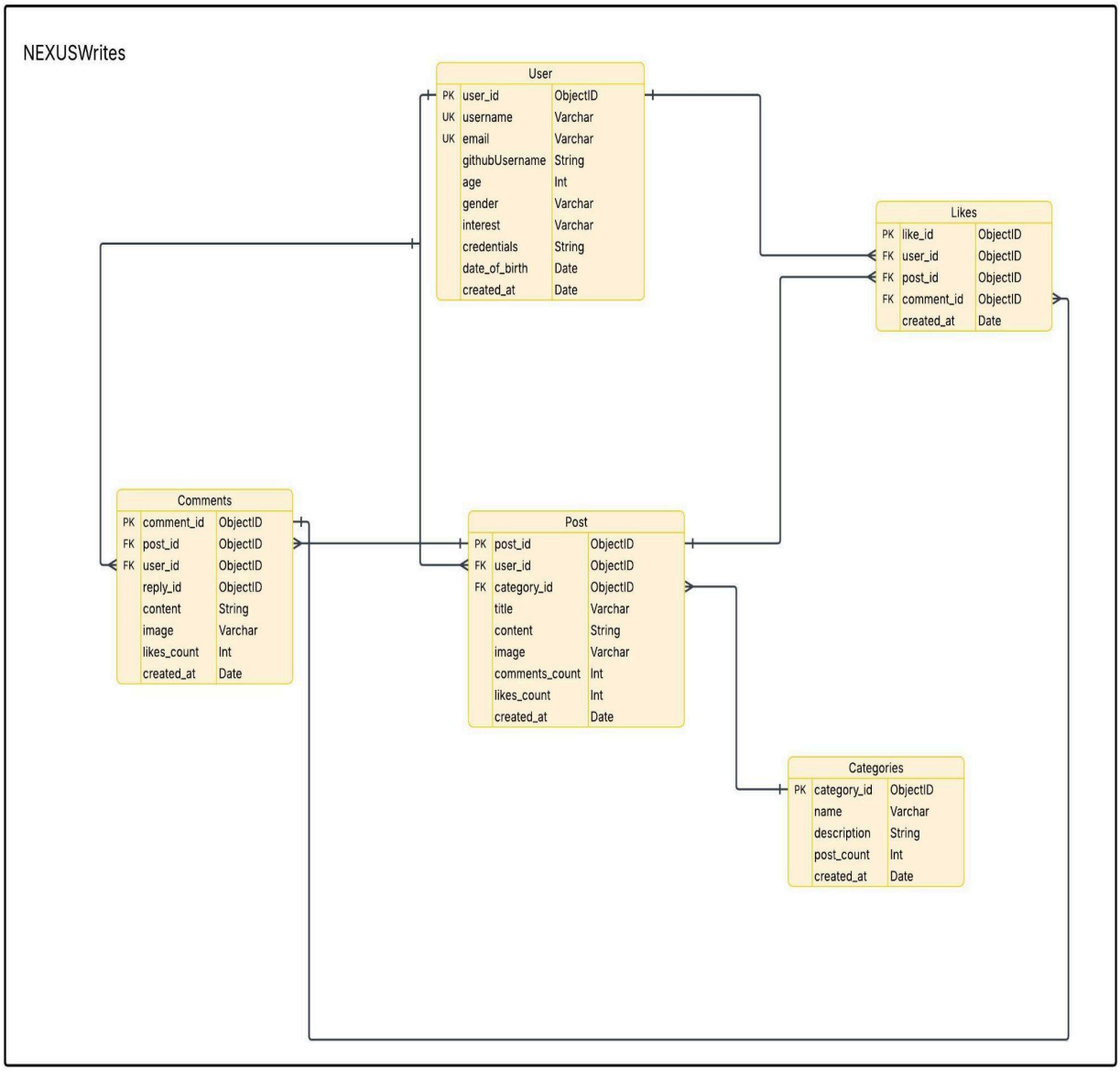
1. **Tech-Centric CRUD & Cloud-Integrated Media:** A primary dashboard for managing technical content. To ensure high performance, the system uses Cloudinary for image hosting. By storing only optimized URL strings in the MongoDB database, the platform displays high-resolution diagrams and screenshots without slowing down the application or bloating server storage.

2. **Live GitHub Portfolio Integration:** To bridge the gap between documentation and real-world code, the platform utilizes a "Read-Only" integration via the GitHub REST API. By providing a GitHub username, the system dynamically fetches and displays a user's public repositories and contributions directly on their NexusWrites profile, turning their blog into a verified technical resume.
3. **Age-Gated Secure Authentication:** A robust security layer featuring a mandatory Date of Birth (DOB) check during registration. The system calculates age server-side to ensure community standards. User security is prioritized by protecting passwords with Bcrypt.js hashing and managing sessions via JSON Web Tokens (JWT).
4. **Hierarchical Discussion System:** To foster professional networking, the platform includes a "Nested" threaded comment system. This allows for organized technical debates and troubleshooting where replies are logically grouped under specific parent comments, making it easy for users to follow complex solution paths.
5. **Dynamic Technical Categorization:** A specialized filtering system that requires users to tag posts under specific IT fields (e.g., CyberSecurity, Web Systems, AI). This allows for a structured "Explore" page where users can filter content by tech stack, ensuring that the platform remains an organized documentation hub rather than a generic blog.

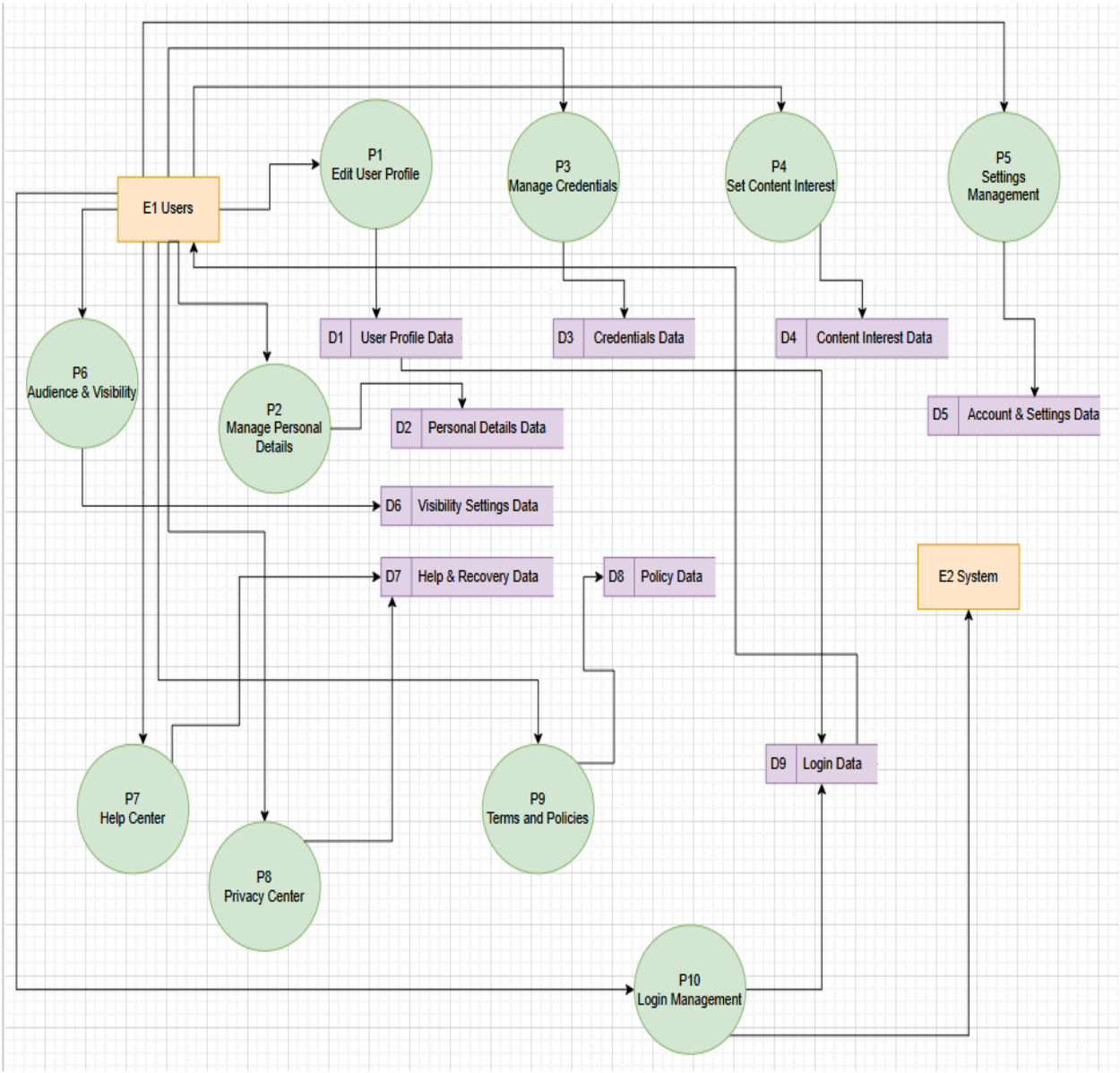
Use Case Diagram



Entity Diagram (ERD)



Data Flow Diagram (LEVEL 1)



LEGEND:

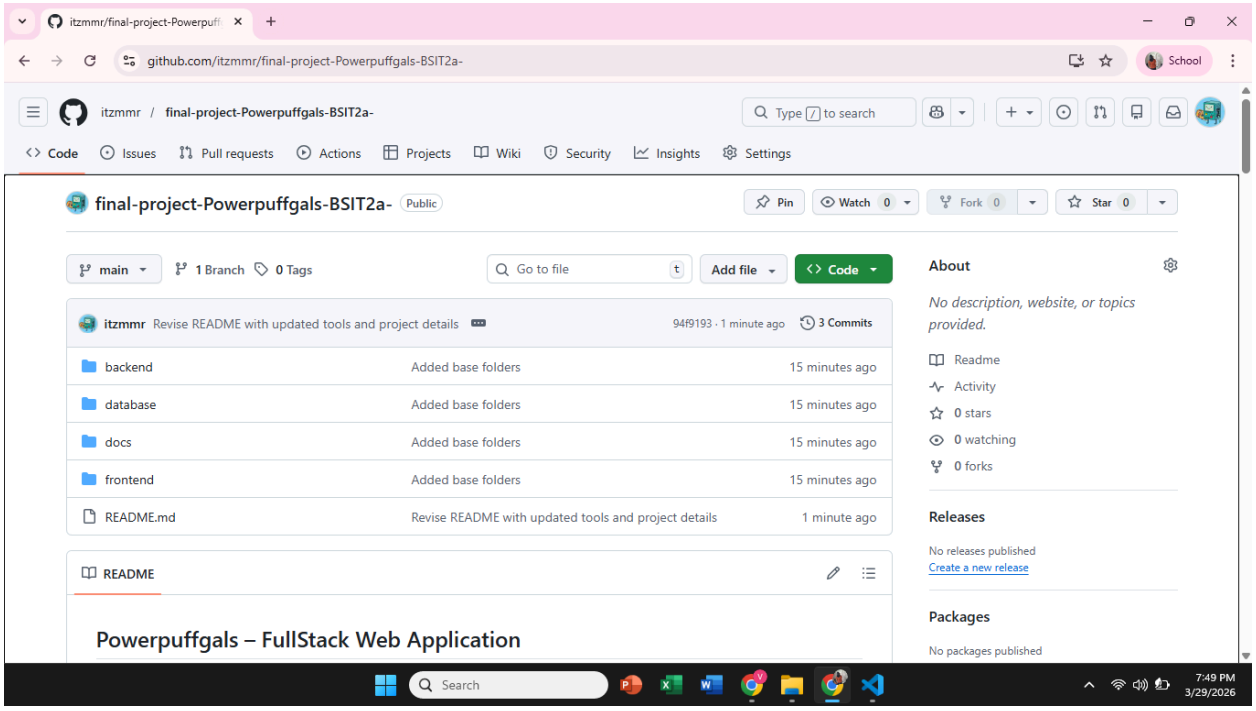
- Processes

- External Entities

- Data Stores

GitHub Repository URL & Screenshot of Repo Structure

URL: https://github.com/itzmmr/final-project-Powerpuffgals-BSIT2a-.git



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